

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

The logs show a multiple network TCP SYN packet that has occurred for over the span of 30 seconds from the source IP 203.0.113.0. This event could be a Denial of Service (DoS) TCP SYN Flooding attack. A type of attack where multiple packets are sent to the server to disrupt its services and functionality. The website is likely receiving a connection timeout due to the packet flooding

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol.

1. The visitor will send an synchronization (SYN) request to the server
2. The server will sent back a synchronization & acknowledgement(SYN/ACK) to the visitor
3. The visitor will send an acknowledgement(ACK) back to the server. The TCP connection is established.

Once the data transfer is done, the Visitor will send a finish request (FIN) and the Server will acknowledge it(ACK) then vice versa.

When a malicious actor tries to send a large number of SYN packet requests all at once, the server will not be able to handle the requests with all the ports open waiting for acknowledgement from the attacker. Thus, overloading the services and ports of the server

The logs indicates that after the attacker sent multiple SYN requests, the server crashed because the server were unable to handle all the SYN request being sent